

Osaid Khan

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Dear Hiring Team,

It is with great pleasure that I submit my candidacy for the **System Development Engineer** role with **Amazon Web Services (AWS)**. I bring over three years of professional experience building distributed backend systems, automation pipelines, and production-grade infrastructure, along with a strong academic foundation as a Master of Science in Computer Science graduate from Pace University, NY (GPA 4.0/4.0).

Through my engineering roles at Sperse and Qualitest, I have built and delivered production systems that directly reflect the development of common libraries, reusable modular components, and automated tooling that this position demands. At Sperse, I designed a Python-based MCP server using FastMCP that exposed 160+ programmatically callable backend actions across multi-tenant environments as a standardized, reusable modular layer — with built-in OpenTelemetry instrumentation and Pydantic validation enforcing reliability at every interface boundary. To simplify and accelerate the software deployment cycle, I built GitHub Actions CI/CD pipelines integrating OpenTelemetry trace-based evaluation suites that gate every production release by validating latency, accuracy, and safety thresholds — mirroring the focus on accelerating build cycles and improving release quality described in this role. I also engineered a distributed multi-agent orchestration platform in Python, LangGraph, and FastAPI routing requests across 40+ specialized sub-agents with cross-functional tenant-aware access control, serving 10,000+ users — work that demanded the same technical trade-offs between short-term delivery goals and long-term architectural scalability that characterize the System Development Engineer responsibilities. Prior to Sperse, at Qualitest, I orchestrated long-running fault-tolerant jobs with AWS Step Functions and Lambda to eliminate timeout failures and improve error recovery across 100K+ weekly distributed executions. I also designed a cloud-agnostic Storage Service in Java and Spring Boot — handling file operations across AWS S3, Google Cloud Storage, and MinIO — which deepened my hands-on systems engineering experience in storage and networking across heterogeneous cloud backends, directly aligning with the systems engineering fundamentals this role requires.

My project work reinforces this foundation in reliability-focused distributed systems. I built Dark Factory, a distributed coding-agent orchestrator that decomposes large requests into dependency-aware task DAGs and executes them across multi-machine agents with retries and timeout handling — applying the same modular, standardized agent interfaces and fault-tolerant execution patterns that are central to developing data center controls products at AWS scale. My graduate coursework at Pace University has further strengthened my ability to reason about system design, performance trade-offs, and architectural decisions with rigor — complementing the practical systems engineering experience I bring from production engineering roles.

I have always been passionate about customer-centric innovation and democratizing access to powerful cloud infrastructure, and I envision myself contributing my talent to the furthering of delivering seemingly infinite, secure, and reliable cloud capacity to customers worldwide. I would love to discuss my interest in this position with your team and can be reached at +1 (945) 304-5781 or khanosaid726@gmail.com. Thank you for your time and consideration, I look forward to connecting!

Sincerely,
Osaid Khan